Thanh Nguyen **Last update: July 21, 2026**

CONTACT INFORMATION

1 Hong Lac Str.
Tan Binh,
Ho Chi Minh City, Vietnam
Tel: (+84)396 939 805

Homepage: www.sitesgoogle.com
Linkedin: www.linkedin.com
✉ E-mail: www.gmail.com

RESEARCH INTERESTS

- My research lies at the intersection of mathematical machine learning, generative modeling, and optimal transport. I am particularly interested in the theoretical foundations of diffusion, score-based, and flow-based generative models, including objective design, numerical discretization, sampling efficiency, robustness, and connections to optimal transport and Schrödinger bridges.

EDUCATION

Ho Chi Minh University of Technology - VNU HCMC, Vietnam 2024–2026

- M.Res. in Applied Mathematics; GPA: **3.5/4.0** (approx. **8.42/10**).
- Thesis: *Optimizing Score Flow Matching for Generative Models Based on Density Estimation.*
- Advisor: Dr. Le Xuan Dai.

Dalat University, Vietnam 2024–2026

- B.Sc. in Mathematics Education; GPA: **3.2/4.0**.

University of Economics Ho Chi Minh City, Vietnam 2020–2023

- B.Sc. in Finance & Banking; GPA: **3.2/4.0**.

RESEARCH EXPERIENCE

- **Graduate Research Assistant - Dr. Le Xuan Dai** Mar 2025–Present
  Department of Applied Mathematics, Ho Chi Minh University of Technology - VNU HCMC.
  Project: Density-Estimation-Based Objective Design for Generative Models.
  - Investigate the theoretical and empirical properties of training objectives for diffusion models, normalizing flows, and variational autoencoders.
  - Develop a density-estimation-based formulation connecting KL minimization with score-matching and flow-matching objectives.
  - Design comparative experiments to evaluate convergence, training stability, and sample quality, and communicate the findings through research reports and visual analyses.
- **Graduate Research Assistant - Dr. Le Xuan Dai** Oct 2024–Mar 2025
  Department of Applied Mathematics, Ho Chi Minh University of Technology - VNU HCMC.
  Project: Bayesian Optimization Algorithms for Machine Learning Models.
  - Studied Bayesian optimization methods and their applicability to hyperparameter and architecture optimization in deep learning.
  - Analyzed acquisition strategies, training dynamics, and model characteristics to identify limitations of conventional Bayesian optimization in high-dimensional settings.
  - Evaluated candidate algorithmic improvements on benchmark datasets and documented the methodology and findings in research reports.

PUBLICATIONS & MANUSCRIPTS

- **Thanh Nguyen** et al. "Optimization of the Objective Function for Generative Models Based on Distribution Density Estimation." *HCMUE Journal of Science*, 2025.
- **Thanh Nguyen** et al. "Improving the Score Flow Matching Objective Function via Optimal Transport." *HUFLIT Journal of Science*, 2026.
- **Thanh Nguyen** et al. "Ranking Objects Based on Multiplicative Consistency of Comparison Matrices: Logarithmic Minimax (Chebyshev) Model and Applications." *HCMUE Journal of Science*, 2026. (*Under-review*).

SKILLS

- *Programming:* Python (PyTorch, NumPy, SciPy, scikit-learn, Matplotlib), R.
- *Mathematical and ML Foundations:* Probability, mathematical statistics, optimization, ODEs, numerical methods, mathematical modeling and simulation; research exposure to score/flow matching, optimal transport, and generative modeling.
- *Research Tools:* Git, Linux/Unix, LaTeX, Jupyter Notebook, manuscript preparation, experiment tracking and visualization.

Academic Programs

- **2025 Compute Ontario Summer School** Jun 02–20, 2025, Virtual/Online.
- **Summer School on Bayesian Models, Computation, and Applications.** Jul 23–Aug 01, 2025, Ho Chi Minh City, Vietnam.
- **The 7th National Conference "Prob-Stats: Research, Application and Teaching".** Aug 04–08, 2025, Quy Nhon, Vietnam.
- **Spring School on Statistics and Machine Learning.** Apr 08–18, 2025, HCMUS - VNU HCMC, Vietnam.
- **Summer School AI Agents.** Sep 08–12, 2025, HUST, Hanoi, Vietnam.

Additional Experience

- **International Mathematics Tutor** 2021–Present
  - Provide individualized instruction in mathematics for IGCSE, GCSE, IB, and A-Level students.
  - Coach students for international mathematics competitions, including TIMO, HKIMO, SASMO, SEAMO, AMC, and ASMO.
- **Data Scientist Intern, PVCFC - PetroVietnam** Oct 2024–Apr 2025
  - Extracted and cleaned operational and sales data from ERP systems to improve data quality and consistency.
  - Built Power BI dashboards for revenue, production, and sales monitoring, supporting data-driven decisions and reducing excess inventory by approximately 5%.

Professional Certificates

- IBM Machine Learning Professional Certificate - IBM.
- IBM Data Science Professional Certificate - IBM.
- Machine Learning / Deep Learning Specialization - DeepLearning.AI.
- Mathematics for Machine Learning and Data Science - DeepLearning.AI.

Service

- Volunteer activities with NGOs and Ivolunteer, including seasonal blood donation drives and community meal support.

References

Dr. Le Xuan Dai
Department of Applied Mathematics, Ho Chi Minh University of Technology - VNU HCMC
E-mail: ytkadai@hcmut.edu.vn

Assoc. Prof. Dr. Huynh Thi Hong Diem
Department of Applied Mathematics, Ho Chi Minh University of Technology - VNU HCMC
E-mail: hthdiem@hcmut.edu.vn